

C-DAC — ASSIGNMENT 02 ANSWERS

QUESTION 01

Electricity Bill Calculation — if-else-if. Read units in main(). Pass to static method calculateBill(). (Slabs: 0-100: ■2, 101-200: ■3, 201-300: ■5, Above 300: ■7).

LOGIC

Read `units` from console in `main()` using `sc.nextInt()`. Pass to static method `calculateBill()` to compute slab charges.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 02 - Question 01: Electricity Bill Calculation
import java.util.Scanner;

public class Q01 {
    static double calculateBill(int units) {
        double bill = 0.0;
        if (units <= 100) {
            bill = units * 2.0;
        } else if (units <= 200) {
            bill = (100 * 2.0) + ((units - 100) * 3.0);
        } else if (units <= 300) {
            bill = (100 * 2.0) + (100 * 3.0) + ((units - 200) * 5.0);
        } else {
            bill = (100 * 2.0) + (100 * 3.0) + (100 * 5.0) + ((units - 300) * 7.0);
        }
        return bill;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter electricity units consumed: ");
        int units = sc.nextInt();

        double totalBill = calculateBill(units);

        System.out.println();
        System.out.println("--- Electricity Bill ---");
        System.out.println("Units Consumed: " + units);
        System.out.println("Total Electricity Bill: Rs. " + totalBill);

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

```
Enter electricity units consumed:
--- Electricity Bill ---
Units Consumed: 250
Total Electricity Bill: Rs. 750.0
```

QUESTION 02

Student Grade Calculation — if-else-if. Read marks in main(). Pass to static method calculateGrade(). (90-100: A, 75-89: B, 60-74: C, 50-59: D, Below 50: F).

LOGIC

Read `marks` from console in `main()` using `sc.nextInt()`. Pass to static method `calculateGrade()` to return grade char.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 02 - Question 02: Student Grade Calculation
import java.util.Scanner;

public class Q02 {
    static char calculateGrade(int marks) {
        if (marks >= 90 && marks <= 100) {
            return 'A';
        } else if (marks >= 75) {
            return 'B';
        } else if (marks >= 60) {
            return 'C';
        } else if (marks >= 50) {
            return 'D';
        } else {
            return 'F';
        }
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter student marks (0-100): ");
        int marks = sc.nextInt();

        char grade = calculateGrade(marks);

        System.out.println();
        System.out.println("--- Result ---");
        System.out.println("Marks: " + marks);
        System.out.println("Calculated Grade: " + grade);

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

```
Enter student marks (0-100):
--- Result ---
Marks: 82
Calculated Grade: B
```

QUESTION 03

Product Discount — if-else-if. Read productId, productName, price in main(). Pass price to static method calculateFinalPrice().

LOGIC

Read `productId`, `productName`, and `price` from console in `main()`. Pass price to static method `calculateFinalPrice()` to apply discount.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 02 - Question 03: Product Discount
import java.util.Scanner;

public class Q03 {
    static double calculateFinalPrice(double price) {
        double discount = 0.0;
        if (price >= 10000) {
            discount = price * 0.20;
        } else if (price >= 5000) {
            discount = price * 0.10;
        } else if (price >= 2000) {
            discount = price * 0.05;
        } else {
            discount = 0.0;
        }
        return price - discount;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Product ID: ");
        int productId = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Product Name: ");
        String productName = sc.nextLine();

        System.out.print("Enter Product Price: ");
        double price = sc.nextDouble();

        double finalPrice = calculateFinalPrice(price);

        System.out.println();
        System.out.println("--- Product Invoice ---");
        System.out.println("Product ID: " + productId);
        System.out.println("Product Name: " + productName);
        System.out.println("Original Price: Rs. " + price);
        System.out.println("Final Price after Discount: Rs. " + finalPrice);

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

Enter Product ID: Enter Product Name: Enter Product Price:

--- Product Invoice ---

Product ID: 101

Product Name: Laptop

Original Price: Rs. 12000.0

Final Price after Discount: Rs. 9600.0

QUESTION 04

Largest of Two Numbers — if-else. Read two integers in main(). Pass to static method findLargest().

LOGIC

Read two numbers from console in `main()`. Pass to static method `findLargest()` and display result.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 02 - Question 04: Largest of Two Numbers
import java.util.Scanner;

public class Q04 {
    static int findLargest(int a, int b) {
        if (a >= b) {
            return a;
        } else {
            return b;
        }
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first integer: ");
        int num1 = sc.nextInt();

        System.out.print("Enter second integer: ");
        int num2 = sc.nextInt();

        int largest = findLargest(num1, num2);

        System.out.println();
        System.out.println("--- Comparison Result ---");
        System.out.println("Number 1: " + num1);
        System.out.println("Number 2: " + num2);
        System.out.println("Largest Number: " + largest);

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

```
Enter first integer: Enter second integer:
--- Comparison Result ---
Number 1: 45
Number 2: 89
Largest Number: 89
```

QUESTION 05

Check Voting Eligibility — if. Read age in main(). Pass to static method isEligible() returning boolean.

LOGIC

Read `age` from console in `main()`. Pass to static boolean method `isEligible()` and print eligibility.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 02 - Question 05: Check Voting Eligibility
import java.util.Scanner;

public class Q05 {
    static boolean isEligible(int age) {
        return age >= 18;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter age: ");
        int age = sc.nextInt();

        boolean eligible = isEligible(age);

        System.out.println();
        System.out.println("--- Voting Status ---");
        System.out.println("Age: " + age);
        if (eligible) {
            System.out.println("Status: Eligible to vote.");
        } else {
            System.out.println("Status: Not eligible to vote.");
        }

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

```
Enter age:
--- Voting Status ---
Age: 21
Status: Eligible to vote.
```

QUESTION 06

Employee Salary Calculation — if-else. Read basic salary in main(). If $\geq$ 50,000 add 20% HRA, else 10% HRA via static method calculateSalary().

LOGIC

Read `basicSalary` from console in `main()`. Pass to static method `calculateSalary()` to add HRA percentage based on salary slab.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 02 - Question 06: Employee Salary Calculation
import java.util.Scanner;

public class Q06 {
    static double calculateSalary(double basicSalary) {
        double hra;
        if (basicSalary >= 50000) {
            hra = basicSalary * 0.20;
        } else {
            hra = basicSalary * 0.10;
        }
        return basicSalary + hra;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter basic salary: ");
        double basicSalary = sc.nextDouble();

        double finalSalary = calculateSalary(basicSalary);

        System.out.println();
        System.out.println("--- Salary Slip ---");
        System.out.println("Basic Salary: Rs. " + basicSalary);
        System.out.println("Final Salary (with HRA): Rs. " + finalSalary);

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

```
Enter basic salary:
--- Salary Slip ---
Basic Salary: Rs. 55000.0
Final Salary (with HRA): Rs. 66000.0
```

QUESTION 07

Temperature Conversion — Operators + if. Convert Celsius to Fahrenheit via static method `convertTemperature()`. In `main()`, check if $> 100^{\circ}\text{F}$.

LOGIC

Read ``celsius`` from console in ``main()``. Pass to static method ``convertTemperature()``. Check if converted Fahrenheit $> 100^{\circ}\text{F}$.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 02 - Question 07: Temperature Conversion
import java.util.Scanner;

public class Q07 {
    static double convertTemperature(double celsius) {
        return (celsius * 9.0 / 5.0) + 32.0;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter temperature in Celsius: ");
        double celsius = sc.nextDouble();

        double fahrenheit = convertTemperature(celsius);

        System.out.println();
        System.out.println("--- Temperature Status ---");
        System.out.println("Celsius: " + celsius + " deg C");
        System.out.println("Fahrenheit: " + fahrenheit + " deg F");

        if (fahrenheit > 100) {
            System.out.println("Notice: Temperature is above 100 deg F!");
        } else {
            System.out.println("Notice: Temperature is within normal range.");
        }

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

```
Enter temperature in Celsius:
--- Temperature Status ---
Celsius: 38.5 deg C
Fahrenheit: 101.3 deg F
Notice: Temperature is above 100 deg F!
```


QUESTION 08

Simple Calculator — if-else-if. Read numbers and operator (+, -, *, /) in main(). Pass to static method calculate(). Handle division by zero.

LOGIC

Read two numbers and operator char from console in `main()`. Pass to static method `calculate()` using `switch` logic.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 02 - Question 08: Simple Calculator
import java.util.Scanner;

public class Q08 {
    static double calculate(double a, double b, char op) {
        switch (op) {
            case '+': return a + b;
            case '-': return a - b;
            case '*': return a * b;
            case '/':
                if (b != 0) return a / b;
                else {
                    System.out.println("Error: Division by zero.");
                    return Double.NaN;
                }
            default:
                System.out.println("Invalid Operator!");
                return Double.NaN;
        }
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        double num1 = sc.nextDouble();

        System.out.print("Enter second number: ");
        double num2 = sc.nextDouble();

        System.out.print("Enter operator (+, -, *, /): ");
        char operator = sc.next().charAt(0);

        double result = calculate(num1, num2, operator);

        System.out.println();
        System.out.println("--- Calculation Result ---");
        System.out.println(num1 + " " + operator + " " + num2 + " = " + result);

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

```
Enter first number: Enter second number: Enter operator (+, -, *, /):
--- Calculation Result ---
25.0 * 5.0 = 125.0
```

QUESTION 09

Bus Ticket Fare — if-else-if. Read name & age in main(). Pass age to static method calculateFare().

LOGIC

Read passenger name and age from console in `main()`. Pass age to static method `calculateFare()` to determine ticket fare.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 02 - Question 09: Bus Ticket Fare
import java.util.Scanner;

public class Q09 {
    static double calculateFare(int age) {
        if (age < 5) {
            return 0.0;
        } else if (age <= 12) {
            return 20.0;
        } else if (age <= 59) {
            return 40.0;
        } else {
            return 25.0;
        }
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter passenger name: ");
        String passengerName = sc.nextLine();

        System.out.print("Enter passenger age: ");
        int age = sc.nextInt();

        double fare = calculateFare(age);

        System.out.println();
        System.out.println("--- Bus Ticket Receipt ---");
        System.out.println("Passenger Name: " + passengerName);
        System.out.println("Passenger Age: " + age);
        System.out.println("Ticket Fare: Rs. " + fare);

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

```
Enter passenger name: Enter passenger age:
--- Bus Ticket Receipt ---
Passenger Name: Ramesh Kumar
Passenger Age: 65
Ticket Fare: Rs. 25.0
```

QUESTION 10

Mobile Data Usage — if-else-if. Read mobile number and GB usage in main(). Pass usage to static method calculateCharge().

LOGIC

Read mobile number and data usage in GB from console in `main()`. Pass usage to static method `calculateCharge()`.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 02 - Question 10: Mobile Data Usage
import java.util.Scanner;

public class Q10 {
    static double calculateCharge(double usageGB) {
        if (usageGB <= 1.0) {
            return 50.0;
        } else if (usageGB <= 5.0) {
            return 100.0;
        } else if (usageGB <= 10.0) {
            return 200.0;
        } else {
            return 350.0;
        }
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter mobile number: ");
        long mobileNumber = sc.nextLong();

        System.out.print("Enter data usage in GB: ");
        double dataUsage = sc.nextDouble();

        double charge = calculateCharge(dataUsage);

        System.out.println();
        System.out.println("--- Data Usage Bill ---");
        System.out.println("Mobile Number: " + mobileNumber);
        System.out.println("Data Usage: " + dataUsage + " GB");
        System.out.println("Total Charge: Rs. " + charge);

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

```
Enter mobile number: Enter data usage in GB:
--- Data Usage Bill ---
Mobile Number: 9876543210
Data Usage: 7.5 GB
Total Charge: Rs. 200.0
```